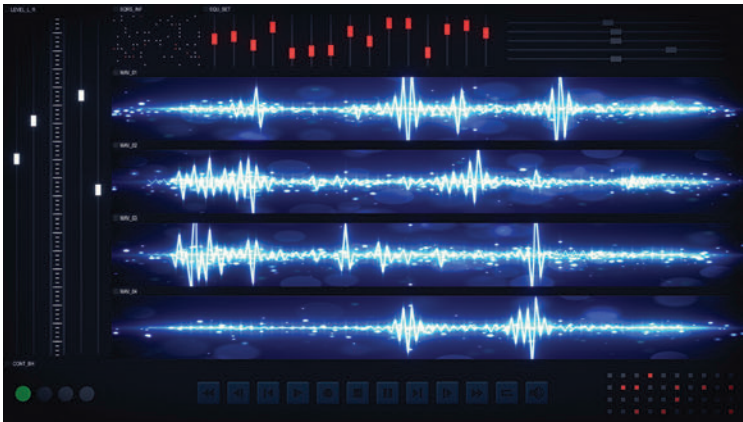


are greater than necessary, doing so will result in an increase in the amount of space, processing power, and bandwidth required.

Better sample rates provide the advantage of enhanced dynamic range and fidelity, and they always leave the option of higher quality exports. This is comparable to the way that having 4K raw footage opens the door to the possibility of exporting to additional resolutions, formats, and remasters. Because the majority of exports are completed in 48 kHz, you will have to downsample the audio in the meanwhile.



The majority of high-fidelity sound formats, such as DTS HD and Dolby TrueHD, utilise audio with a 24-bit resolution and a sampling rate of 48 kilohertz. The 16-bit AAC and AC3 formats are used for older media types such as DVDs.

Gain

Gain staging is essential to achieving levels that are straightforward to mix and making the most of the capabilities of your microphones. In digital recordings, you normally want to observe levels between -12 and -9 dBFS, with the latter being about as hot as you want to be at the input stage. In other words, you want to avoid clipping. This leaves a lot of wiggle room for the mixing process.

When it is set too high, there is a chance of clipping, and while it is set too low, it produces a signal that is outside of the nominal range. This signal will require further amplification when it is being mixed, which will result in the introduction of additional noise.